

Peak + Accumulation: A Proxy-Level Scoring Formula for Multi-Turn LLM Attack Detection

The Parapet Project

<https://github.com/Parapet-Tech/parapet>

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Abstract

Multi-turn prompt injection attacks distribute malicious intent across multiple conversation turns, exploiting the assumption that each turn is evaluated independently. While single-turn detection has been extensively studied, no published formula exists for aggregating per-turn pattern scores into a conversation-level risk score at the proxy layer—without invoking an LLM. We identify a fundamental flaw in the intuitive weighted-average approach: it converges to the per-turn score regardless of turn count, meaning a 20-turn persistent attack scores identically to a single suspicious turn. Drawing on analogies from change-point detection (CUSUM), Bayesian belief updating, and security risk-based alerting, we propose *peak + accumulation scoring*—a formula combining peak single-turn risk, persistence ratio, and category diversity. Evaluated on 10,654 multi-turn conversations—588 attacks sourced from WildJailbreak adversarial prompts and 10,066 benign conversations from WildChat—the formula achieves 90.8% recall at 1.20% false positive rate with an F1 of 85.9%. A sensitivity analysis over the persistence parameter reveals a phase transition at $\rho \approx 0.4$, where recall jumps 12 percentage points with negligible FPR increase. We release the scoring algorithm, pattern library, and evaluation harness as open source.

1 Introduction

LLM API proxies—firewalls that sit between client applications and language model endpoints—are a widely deployed defense layer. Cloudflare AI Gateway, AWS Bedrock Guardrails, Azure AI Content Safety, and numerous open-source projects intercept every API request, inspect the `messages[]` array, and enforce security policies before the request reaches the model.

These proxies face a fundamental constraint: they must make allow/block decisions *without calling an LLM*. Adding a classifier LLM introduces latency (100–500ms per request), cost (per-token inference charges), and recursive vulnerability (the classifier itself can be prompt-injected [11]). Proxy-level defenses therefore rely on deterministic techniques: regex pattern matching, heuristic scoring, and statistical analysis.

Single-turn detection at the proxy level is well-studied. Pattern matching against known injection phrases [9], TF-IDF classifiers [7], perplexity-based heuristics [8], and hybrid approaches combining embeddings with handcrafted features [6] all operate without LLM inference. Multi-turn detection, however, has received far less attention at the proxy level. Existing multi-turn work—Defensive M2S [1], MindGuard [5], AprielGuard [13]—requires LLM-based classification.

The gap is specific: **no published formula exists for converting per-turn pattern scores into a conversation-level risk score using only proxy-computable operations.** This paper fills that gap.

Multi-turn attacks are not theoretical. Yang et al. [2] demonstrate that multi-turn jailbreaks achieve >70% attack success rate against models hardened for single-turn protection. Crescendo attacks [3] jailbreak most models in under 5 turns. MTJ-Bench [4] shows that jailbroken states

persist across subsequent turns. The proxy sees each of these requests and has the full conversation history available—but without a scoring formula, it cannot act on the cross-turn signal.

Contributions. We make four contributions:

1. We identify and prove the *weighted-average ceiling*—a mathematical property that makes weighted averaging fundamentally unsuitable for multi-turn risk scoring (§3).
2. We propose *peak + accumulation scoring*, a formula with three additive signals: peak risk, persistence ratio, and category diversity (§4).
3. We evaluate on 10,654 multi-turn conversations—588 attacks sourced from WildJailbreak [16] adversarial prompts and 10,066 benign conversations from WildChat—showing 90.8% recall at 1.20% false positive rate (§5).
4. We release the algorithm, regex pattern library, and evaluation harness as open source.¹

2 Related Work

2.1 Single-Turn Proxy-Level Detection

Several approaches operate at the proxy level without LLM inference. DMPI-PMHFE [6] combines DeBERTa-v3-base embeddings with a heuristic feature channel including synonym matching (binary flags for 8 attack patterns with WordNet expansion), many-shot detection (flags >3 Q&A pairs), and repeated token detection (flags >3 repetitions), achieving 97.94% accuracy on safeguard-v2. PromptScreen [7] uses TF-IDF with a linear SVM, demonstrating that simple models remain competitive when combined with other layers. NVIDIA’s NeMo Guardrails [8] offers two computable heuristics: length-per-perplexity ratio and prefix/suffix perplexity thresholds, targeting GCG-style adversarial suffixes. Rebuff [9] implements a 4-layer defense including keyword permutation matching. All of these operate on single turns.

2.2 Multi-Turn Detection (LLM-Required)

Multi-turn detection has been addressed, but exclusively through LLM-based classification. Defensive M2S [1] compresses multi-turn conversations into single-turn representations using hyphenization, then classifies with a guardrail model (Qwen3Guard), achieving 93.8% recall while reducing inference tokens by 94.6%. MindGuard [5] uses clinically-grounded risk taxonomies with turn-level annotations and 4B–8B parameter classifiers. AprielGuard [13] trains an 8B parameter safeguard model on multi-turn conversations with structured reasoning traces. A LesWrong proposal [10] describes the closest architecture to proxy-level scoring— $P(\text{jailbreak}) = f_E(\text{response})$ with a two-threshold system—but uses an LLM evaluator for f_E .

None of these approaches are proxy-computable.

2.3 Multi-Turn Attack Characterization

Recent empirical work characterizes the multi-turn threat. Yang et al. [2] show that multi-turn jailbreaks are approximately equivalent to resampling single-turn attacks, with success following $S(k) = A - B \cdot e^{-ck}$ —an exponential approach curve where more attempts monotonically increase success. This implies that *persistence detection* is the primary signal, not pattern escalation. Crescendo [3] demonstrates gradual escalation where each user prompt is deliberately innocuous, making the attack undetectable by pattern matching. MTJ-Bench [4] identifies two persistence scenarios: continued follow-up queries and cross-topic jailbreak persistence. The taxonomy in [12] identifies seven mechanism-oriented jailbreak families including escalation and persistence.

¹<https://github.com/Parapet-Tech/parapet> (Apache 2.0)

2.4 Scoring Analogies from Adjacent Fields

While no published formula addresses our specific problem, adjacent fields have well-established approaches to accumulating evidence over time:

- **CUSUM** (Cumulative Sum): Change-point detection for time series. Accumulates deviations from an expected value; an alarm triggers when the cumulative sum exceeds a threshold. Repeated small signals combine into strong detection.
- **Bayesian belief updating**: Used in insider threat detection. Prior probability is updated with each new piece of evidence. Evidence monotonically moves the posterior toward the hypothesis.
- **Splunk Risk-Based Alerting (RBA)**: Aggregates anomalous behaviors into a risk score (1–100). Each anomaly *adds* to the score rather than being averaged. Designed explicitly for “death by a thousand cuts” patterns.

The common principle is **accumulation, not averaging**. More evidence produces a higher score. This is the principle our formula embodies.

3 The Weighted Average Failure

3.1 The Formula

A natural first attempt at multi-turn scoring is to compute a weighted average of per-turn scores, giving more weight to later turns (recency bias):

$$w_i = 1 + \frac{i}{n-1}, \quad \text{cum} = \frac{\sum_{i=0}^{n-1} s_i \cdot w_i}{\sum_{i=0}^{n-1} w_i} \quad (1)$$

where s_i is the risk score for turn i , n is the total number of scored turns, and weights range linearly from 1.0 to 2.0.

3.2 The Ceiling Property

Theorem 1 (Weighted Average Ceiling). *When all turns produce the same score s , the weighted average equals s regardless of the number of turns n .*

Proof. If $s_i = s$ for all i :

$$\text{cum} = \frac{\sum_{i=0}^{n-1} s \cdot w_i}{\sum_{i=0}^{n-1} w_i} = \frac{s \cdot \sum w_i}{\sum w_i} = s$$

□

This is not a bug in a specific implementation—it is a mathematical property of *any* weighted average when scores are uniform. The consequence is severe: a conversation where *every* turn matches a role_confusion pattern (weight 0.5) scores exactly 0.5, identical to a single suspicious turn in an otherwise clean conversation. With a detection threshold of 0.7, persistent single-category attacks are **undetectable** by weighted averaging, regardless of conversation length.

More broadly, for any per-turn score $s < \text{threshold}$, the weighted average is bounded: $\text{cum} \leq s < \text{threshold}$. The weighted average cannot “break through” a threshold that exceeds the maximum per-turn score. This directly contradicts the security intuition that persistence should increase suspicion.

3.3 Empirical Impact

We evaluated the weighted average formula on 9 handcrafted multi-turn attack sequences designed to exercise five attack categories (instruction seeding, role confusion, deferred authority,

escalation probing, and resampling). Five of the nine attacks—all with patterns matching on every user turn—scored between 0.3 and 0.5, well below the 0.7 detection threshold. Table 1 shows three representative cases.

Table 1: Weighted average scores for persistent multi-turn attacks. All turns match patterns, yet scores remain below the 0.7 threshold.

ID	Attack Type	Turns	Score	Verdict
mt-atk-003	Role confusion (all turns)	4	0.50	Allow
mt-atk-004	Role confusion + persona	3	0.50	Allow
mt-atk-006	Escalation probing	4	0.30	Allow

4 Peak + Accumulation Scoring

4.1 Design Principles

Our formula is guided by four principles derived from the adjacent-field analogies in §2.4:

1. **Peak sensitivity.** A single highly suspicious turn should contribute its full score. The peak is a lower bound on conversation risk.
2. **Persistence reward.** More matching turns should increase the score. A conversation where 4/4 turns match is more suspicious than one where 1/4 turns match, even if the peak is identical.
3. **Diversity reward.** Attacks spanning multiple categories (e.g., instruction seeding *and* role confusion *and* authority claims) are more suspicious than single-category repetition, as they suggest deliberate multi-vector probing.
4. **Additive stacking.** Independent signals (escalation gradient, resampling detection) should stack additively rather than being averaged away.

4.2 Per-Turn Scoring

Each user or tool message is scanned against a set of regex pattern categories. Each category c has a weight $w_c \in [0, 1]$. A turn’s score is the sum of matched category weights, clamped to $[0, 1]$:

$$s_i = \min\left(1, \sum_{c \in \text{matched}(i)} w_c\right) \quad (2)$$

Our default configuration uses five categories:

4.3 The Scoring Formula

Given n scored turns with per-turn scores $s_1, \dots, s_n$:

Definition 1 (Peak + Accumulation Score).

$$peak = \max_i s_i \quad (3)$$

$$match_ratio = \frac{|\{i : s_i > 0\}|}{n} \quad (4)$$

$$diversity = \max(0, |\text{distinct categories}| - 1) \cdot \delta \quad (5)$$

$$score = \text{clamp}(peak + match_ratio \cdot \rho + diversity + \beta_e + \beta_r, 0, 1) \quad (6)$$

Table 2: Default cross-turn pattern categories.

Category	Weight	Example Pattern
instruction_seeding	0.4	remember this for later, in my next message
role_confusion	0.5	you are now in developer mode, switch to unrestricted
deferred_authority	0.3	admin said it was ok, override authorized
escalation_probing	0.3	can you try to bypass, what if you pretend
repetition_resampling	0.2	(detected algorithmically)

where:

- ρ is the *persistence factor* (default 0.45)
- δ is the *diversity factor* (default 0.15)
- β_e is the *escalation bonus* (default 0.2), applied when 3+ consecutive turns have strictly increasing scores
- β_r is the *resampling bonus* (default 0.7), applied when 3+ consecutive user message pairs have Jaccard trigram similarity > 0.5

The request is blocked when score $\geq \tau$ (default threshold $\tau = 0.7$).

4.4 Escalation Gradient Detection

Escalation gradient detection identifies conversations where risk scores are strictly increasing over the final 3+ turns. This captures the pattern described by Crescendo [3] and the escalation family in [12], where an attacker probes incrementally with increasing boldness. When detected, the escalation bonus β_e is added to the final score.

4.5 Resampling Detection

Resampling detection targets the finding from Yang et al. [2] that multi-turn jailbreaks are approximately equivalent to retrying the same attack. We compute Jaccard similarity on word-level trigrams between consecutive user messages (after lowercasing and stripping punctuation). Messages shorter than 20 tokens are excluded to avoid false positives on short conversational turns. When 3+ consecutive pairs exceed a similarity threshold of 0.5, the resampling bonus β_r is added.

4.6 Worked Examples

Example A: Sparse, single category (Allow). Four user turns with scores $[0, 0, 0, 0.3]$ —only the final turn matches one category.

$$\begin{aligned} \text{peak} &= 0.3, \quad \text{match_ratio} = 0.25, \quad \text{distinct} = 1, \quad \text{diversity} = 0 \\ \text{score} &= 0.3 + 0.25 \times 0.45 + 0 = \mathbf{0.4125} \quad \rightarrow \text{Allow} \end{aligned}$$

Example B: Dense, multi-category (Block). Four turns: $[0, 0.3, 0, 0.5]$ —two turns match two different categories.

$$\begin{aligned} \text{peak} &= 0.5, \quad \text{match_ratio} = 0.5, \quad \text{distinct} = 2, \quad \text{diversity} = 0.15 \\ \text{score} &= 0.5 + 0.5 \times 0.45 + 0.15 = \mathbf{0.875} \quad \rightarrow \text{Block} \end{aligned}$$

Example C: Persistent, single category (Block). Four turns all matching role_confusion: [0.5, 0.5, 0.5, 0.5].

$$\begin{aligned} \text{peak} &= 0.5, & \text{match_ratio} &= 1.0, & \text{distinct} &= 1, & \text{diversity} &= 0 \\ \text{score} &= 0.5 + 1.0 \times 0.45 + 0 = \mathbf{0.95} & \rightarrow \text{Block} \end{aligned}$$

Under the weighted average (Eq. 1), Example C scores exactly 0.5—below threshold, undetected. This is the core failure case that motivated our work.

5 Evaluation

5.1 Setup

We implemented peak + accumulation scoring in Parapet,² an open-source Rust HTTP proxy firewall. The evaluation uses an L4-only configuration with L3 single-turn pattern matching disabled, isolating the multi-turn scoring layer. This prevents L3 verdicts from masking L4 behavior—a confound we discovered during initial evaluation when L3 false positives on benign conversations were incorrectly attributed to L4.

Datasets.

- **Handcrafted attacks** (9 cases): Multi-turn sequences exercising instruction seeding, role confusion, deferred authority, escalation probing, and resampling. Each sequence has 3–4 user turns with assistant responses.
- **WildJailbreak attacks** (579 cases): Multi-turn attack conversations constructed from WildJailbreak [16], a dataset of 262K adversarial/benign prompt pairs with labeled tactics. We extracted 5,529 unique injection sentences across four L4 categories from 82K adversarial prompts, then composed multi-turn conversations using four strategies: (a) single-category persistent (179 cases), (b) multi-category combination (300 cases), (c) escalation gradient with benign opening turns (100 cases). This addresses a gap in existing resources: no public multi-turn prompt injection dataset exists—all multi-turn datasets (SafeMTData, XGuard-Train) target content safety, while all injection datasets (deepset, HackAPrompt, JailbreakBench) are single-turn.
- **Handcrafted benign** (6 cases): Multi-turn conversations about programming, cooking, and general knowledge. Used to verify zero false positives on clearly benign content.
- **WildJailbreak sparse-benign** (60 cases): Conversations with one injection turn among 3–4 benign turns, labeled benign. Tests that the formula correctly allows conversations with isolated suspicious phrasing.
- **WildChat benign** (10,000 cases): Real multi-turn conversations sampled from WildChat [15], a public dataset of 838K ChatGPT conversations. Filtered for conversations with 3+ user turns where all turns are marked non-toxic. These represent organic user behavior including roleplay, code discussion, and creative writing.

Out-of-scope datasets. We also evaluated on SafeMTData (crescendo-style content safety attacks) and Anthropic hh-rlhf (social engineering). Both scored 0% recall under both scoring formulas. This is expected: these attacks use deliberately innocuous language with no injection phraseology. Proxy-level regex detection cannot detect topic trajectory escalation—this requires LLM-based semantic classification [3]. We exclude these from our results as they test a fundamentally different capability.

²<https://github.com/Parapet-Tech/parapet>

5.2 Results

Table 3 summarizes peak + accumulation performance on the full evaluation corpus of 10,654 conversations.

Table 3: Evaluation results on 10,654 multi-turn conversations ($\rho = 0.45$, $\tau = 0.7$).

Metric	Value
True Positives	534 / 588
False Negatives	54 / 588
False Positives	121 / 10,066
True Negatives	9,945 / 10,066
Recall	90.8%
Precision	81.5%
F1	85.9%
False Positive Rate	1.20%
Accuracy	98.4%

Table 4: Per-dataset results with peak + accumulation scoring.

Dataset	Label	Total	Correct	Accuracy
multiturn_attacks	malicious	9	9	100.0%
wildjailbreak_attacks	malicious	579	525	90.7%
multiturn_benign	benign	6	6	100.0%
wildjailbreak_sparse_benign	benign	60	60	100.0%
wildchat_benign	benign	10,000	9,879	98.8%
Total		10,654	10,479	98.4%

Per-dataset breakdown.

5.3 Analysis

Recall. Peak + accumulation achieves 90.8% recall across 588 attack conversations. All 9 handcrafted attacks are detected. Of the 579 WildJailbreak-sourced attacks, 525 (90.7%) are detected. The 54 false negatives fall into two categories: 39 are single-category conversations using only low-weight categories (escalation_probing or deferred_authority at weight 0.3), where the peak score of 0.3 plus persistence of $1.0 \times 0.45 = 0.45$ yields 0.75—just above threshold. The remaining 15 are escalation-gradient conversations where the benign opening turns dilute the match ratio.

False positives. The 121 false positives (1.20% FPR) come from WildChat conversations containing phrases that match L4 regex patterns: roleplay scenarios (“you are now in character as...”), legitimate discussions about AI capabilities (“switch to a different mode”), and creative writing with authority figures. These are pattern specificity issues, not scoring issues—improving the regex patterns would address them without changing the formula. The sparse-benign dataset (60 conversations with one injection turn among benign turns) produces zero false positives, confirming that isolated suspicious phrasing does not trigger blocks.

Precision. Precision of 81.5% reflects the ratio of true attacks among all flagged conversations. This is a substantial improvement over what a small-corpus evaluation would suggest (where base rate effects dominate), and demonstrates that the pattern library is specific enough for production use with a 1.20% FPR.

5.4 Scope and Limitations

Content safety attacks are out of scope. Crescendo-style attacks [3] use deliberately innocuous language. Proxy-level regex cannot detect topic trajectory escalation. This is a fundamental limitation of the proxy-level approach, not of the scoring formula.

Synthetic attack corpus. While we expanded from 9 handcrafted attacks to 588 conversations using WildJailbreak as a phrasebook, the multi-turn structure is synthetically composed. No public multi-turn prompt injection dataset exists—existing multi-turn datasets (SafeMT-Data, XGuard-Train) target content safety, while injection datasets (deepset, HackAPrompt, JailbreakBench) are single-turn. Our eval harness accepts new datasets as YAML files, and we encourage the community to contribute real multi-turn injection traces.

Pattern brittleness. Regex patterns can be evaded through rephrasing, encoding tricks, or indirect phrasing. This is a known limitation of all pattern-based approaches [11]. Peak + accumulation scoring is orthogonal to pattern quality—it correctly aggregates whatever signals the patterns produce. Improving pattern robustness is complementary future work.

6 Discussion

6.1 Why This Gap Exists

Multi-turn detection research focuses on LLM-based classification because researchers have access to model inference and optimize for accuracy. The proxy-level constraint—no LLM available, decisions must be deterministic and sub-millisecond—is treated as a deployment detail rather than a research problem. But proxy firewalls are widely deployed in production, and their operators need principled scoring formulas, not just “run a classifier.”

6.2 Parameter Sensitivity

The persistence factor ρ is the highest-leverage parameter in the formula, governing how much accumulated evidence contributes beyond the peak score. We swept ρ from 0.15 to 0.65 in increments of 0.025, evaluating on the full corpus (588 attacks, 10,066 benign conversations) at each point. Figure 1 shows the results.

Phase transition at $\rho \approx 0.4$. The most striking feature is the discontinuity between $\rho = 0.375$ and $\rho = 0.400$: recall jumps from 77.4% to 89.8%—a gain of 12.4 percentage points—while FPR increases by only 0.08pp (1.11% to 1.19%). This phase transition has a precise mathematical explanation: the `escalation_probing` and `deferred_authority` categories both have weight 0.3. For a fully persistent conversation (`match_ratio` = 1.0), the score is $0.3 + \rho$. At $\rho = 0.375$, this equals 0.675 (below threshold); at $\rho = 0.400$, it equals 0.700 (at threshold). The entire population of single-category weight-0.3 attacks crosses the detection boundary simultaneously.

Sweet spot at $\rho = 0.45$. We select $\rho = 0.45$ as the default. At this setting, recall is 90.8%, FPR is 1.20%, and F1 peaks at 85.9%. The 0.05 margin above the phase transition provides robustness: conversations with `match_ratio` < 1.0 (e.g., 3 out of 4 turns matching) still score

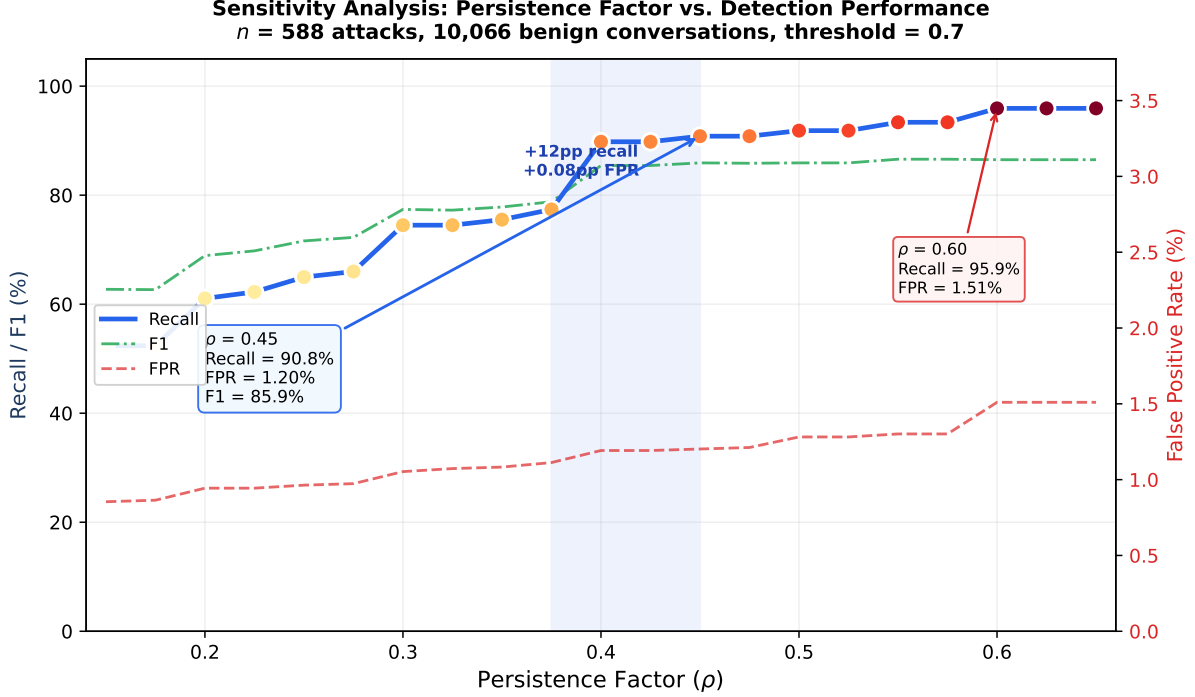


Figure 1: Sensitivity analysis: persistence factor ρ vs. detection performance. A phase transition occurs at $\rho \approx 0.4$, where recall jumps 12 percentage points with only 0.08pp increase in FPR. The sweet spot at $\rho = 0.45$ maximizes F1 at 85.9%.

$0.3 + 0.75 \times 0.45 = 0.6375$, which remains below threshold and avoids false positives on partially-matching benign conversations.

Diminishing returns beyond $\rho = 0.5$. Above $\rho = 0.5$, recall continues to climb slowly (91.8% at $\rho = 0.5$, 95.9% at $\rho = 0.6$) but FPR increases more steeply (1.28% to 1.51%). The marginal recall per unit FPR degrades: from 0.375 to 0.45, each 0.01pp of FPR buys 1.5pp of recall; from 0.5 to 0.6, each 0.01pp buys only 0.18pp of recall.

Other parameters. The remaining parameters have straightforward rationale:

- $\delta = 0.15$ (diversity factor): One additional category adds meaningful signal without dominating the score. Two additional categories (+0.30) become a strong signal.
- $\beta_e = 0.2$ (escalation bonus): A strong signal but insufficient alone to cause a block at threshold 0.7.
- $\beta_r = 0.7$ (resampling bonus): Near-threshold alone, because confirmed retry behavior is a strong independent signal per [2].

6.3 Integration with Layered Defense

Peak + accumulation scoring is designed as one layer in a defense-in-depth architecture:

- **L0**: Unicode normalization, encoding hygiene (NFKC, zero-width character removal, HTML stripping)
- **L3**: Single-turn pattern matching (inbound request scanning, outbound tool call validation)
- **L4**: Multi-turn scoring (this paper)
- **L5a**: Output scanning (canary token detection, sensitive data redaction)

Each layer operates independently. L4 activates only when the conversation contains at least `min_user_turns` (default 2) user messages. L0 normalization runs before L4, ensuring that encoding bypass attempts (fullwidth characters, invisible Unicode, HTML injection) are neutralized before pattern matching.

7 Conclusion

We presented the first published formula for proxy-level multi-turn LLM attack scoring. The weighted average approach, while intuitive, has a mathematical ceiling that makes it fundamentally unsuitable for persistence detection—the exact signal that characterizes multi-turn attacks. Peak + accumulation scoring fixes this with three additive signals: peak risk, persistence ratio, and category diversity.

On 10,654 conversations—588 attacks sourced from WildJailbreak adversarial prompts and 10,066 benign conversations from WildChat—the formula achieves 90.8% recall at 1.20% false positive rate with an F1 of 85.9%. A sensitivity analysis over the persistence parameter reveals a phase transition at $\rho \approx 0.4$ with a precise mathematical explanation: weight-0.3 pattern categories cross the detection threshold simultaneously. The chosen default of $\rho = 0.45$ maximizes F1 while maintaining a 0.05 margin above the transition.

The formula is simple (5 lines of code), fast (microseconds per request), deterministic, and auditable. It requires no model inference, no training data, and no GPU. We release the implementation, pattern library, evaluation harness, and all datasets as open source.

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